The E-Ink Reading Pipeline

source · **github.com/forcewake**

How plain markdown becomes a document worth putting down your phone for — a tour of every stage, from typography arithmetic to halftone plates, exercised by this very document.

> *The reMarkable is the best reading surface in the house and the worst place to render for. Both facts are about the same screen.*

Содержание

Typography tuned to the panel

Every page is exactly 157.7 × 210.3 mm — the reMarkable 2 display at 226 dpi, edge to edge. No fit-to-width scaling, no zoom, no letterboxing: one PDF page is one screen, forever.

The body is a true 12 pt at 16 CSS px, which sounds trivial until you measure what most "send to e-reader" pipelines produce — usually 9–10 pt letterboxed corpses. Margins are 13 mm at the sides and top, 15 mm at the bottom for the thumb. A line settles at ~65 characters: the middle of the classical comfortable range.

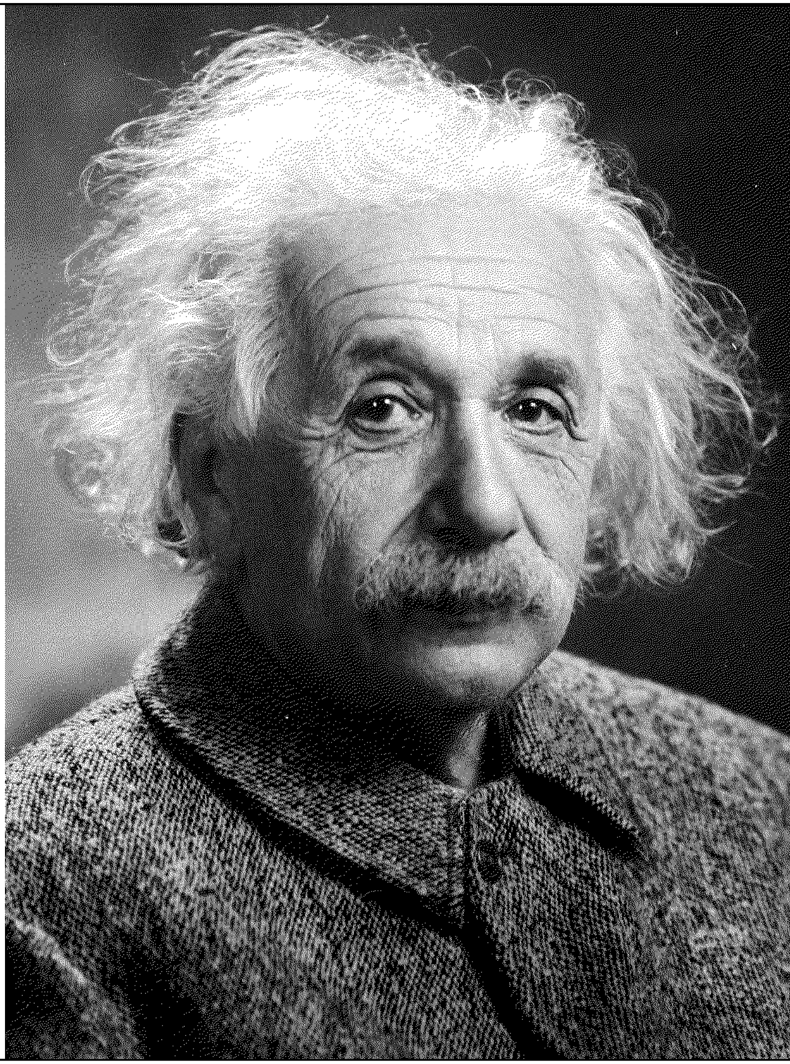
Decision	Value	Why
Page size	157.7 × 210.3 mm	exactly the panel — zero device scaling
Body	16 px = 12 pt	the e-ink legibility floor
Leading	1.55	e-ink needs air between lines
Margins	13 / 15 mm	thumb room, pen rest
Column	single, ~65 chars	PDFs don't reflow

Plates, not pictures

Ada Lovelace, portrait

Photographs never ship raw. Each one is fetched, EXIF-oriented, converted to grayscale, auto-contrasted, and dithered with 1-bit Floyd–Steinberg at the panel's own pixel density — 225 ppi, one dither dot per device pixel. The result reads like a newspaper plate instead of blurry gray mud.

A second plate



Albert Einstein, 1921

A figure shares its page with its heading and text: plates are capped at two thirds of page height, so sections read like magazine spreads instead of orphaned image dumps. Captions come from the alt text, set in mono.

Navigation as a first-class feature

Every render ships with a sidebar outline for the tablet's UI and correct PDF metadata. Documents with four or more sections — this one qualifies — get a clickable table of contents whose page numbers are *measured from the rendered PDF*: render once, find where each section really landed, render again with the truth. Pagination is a fact, not a hope.

Code and structure survive

Sections open their own pages. Code blocks are boxed in mono and never tear across a page break; tables repeat their headers on continuation pages; headings never dangle at the bottom of a page without their content.

```
# the actual pipeline, compressed to its essence
html = grid_css(markdown(article))          # typography +
themes
pdf = chrome(headless).print(html)          # pages exactly
446.88 × 595.92 pt
pages = measure(pdf)                        # where did
sections really land?
pdf = chrome(headless).print(toc(html, pages))  # contents
with real numbers
ship(rmapi.upload(pdf, "/Inbox"))           # cloud → tablet
in ~a minute
```

Themes for the occasion

Three type systems share the same layout engine: `grid` for technical briefs (this document), `book` — serif, hairline rules, chapter openings — for essays you'll read for an hour, and `compact` at 14 pt for manuals that must fit the flight, not the desk.

Ship it

```
remarkable_send.py --file showcase.md --source
github.com/forcewake
```

Markdown in. A folder on the tablet out. That is the whole pipeline.